

ENVS 193DS Final

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Link to GitHub repo [here](#).

Set up

```
# loaded in packages
library(tidyverse)
library(janitor)
library(here)
library(readxl)
library(dplyr)
library(effsize)
library(MuMIn)
library(DHARMA)

# read in data and saved as object called `nest`
nest <- readr::read_csv(here::here("data","nest_data_final.csv"))

# read in data and saved as object called `rxntime`
rxntime <- readr::read_csv(here::here("data","rxntime.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used linear regression. In part 2, they used a chi-square test of independence.

b. Suggestions for rewriting

Part 1. We found that the distance of microhabitat to water edge predicts the likelihood of usage by American Avocets.

Part 2. We found that the relationship between bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat type (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat) is statistically significant.

We found that the prevalence of bird species (American Avocet, Forster's Tern, Black-necked Stilt) differ significantly by habitat type (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

OR

We found that different bird species (American Avocet, Forster's Tern, Black-necked Stilt) show preference for certain habitat types (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

c. Further analysis

Predation pressure can be studied as a discrete variable (count of individuals); the presence of predators could make microhabitats too risky for the birds to frequent safely. As such, the birds may choose to avoid predator territories, reducing the probability of American Avocet microhabitat use in those locations. Predator data may be acquired through methods such as tracking, video sightings/ identification, and/ or direct observation.

categorical or continuous

Problem 2. Data analysis

a. Clean and wrangle

```
# created new object from nest data frame
nests_clean <- nest |>
  # selected columns of interest
  select(height_cm, ant_species, case_control) |>
  # filtered ant species
  filter(
    ant_species == "AB" |
    ant_species == "RRB" |
    ant_species == "BBR") |>
  # reordered the levels of ant species
  mutate(ant_species = factor(ant_species,
                              levels = c("AB",
                                           "RRB",
                                           "BBR"),
```

```

"BBR")))) |>
# created a new column to store ant scientific names
mutate(ant_species = case_when(
  ant_species == "AB" ~ "Crematogaster sjostedti",
  ant_species == "RRB" ~ "Crematogaster mimosae",
  ant_species == "BBR" ~ "Crematogaster nigriceps"))

# display of data in a sliced sample
slice_sample(nests_clean,
  n = 10,
  by = NULL,
  weight_by = NULL,
  replace = FALSE)

```

```

# A tibble: 10 x 3
  height_cm ant_species case_control
  <dbl> <chr> <dbl>
1    130 Crematogaster mimosae         0
2    136. Crematogaster mimosae         0
3    215 Crematogaster mimosae         0
4    134. Crematogaster mimosae         0
5    120. Crematogaster mimosae         0
6    254 Crematogaster mimosae         1
7    212. Crematogaster mimosae         1
8    109 Crematogaster nigriceps         0
9    222. Crematogaster mimosae         0
10   353 Crematogaster mimosae         0

```

b. Response variable

A tree that contained a nest (“used” tree) is denoted with a (1) and trees without nests (“available” trees) are denoted with a (0).

c. Purpose of study

Tree height is a continuous variable, whereas ant species is a categorical variable. Trees that are taller may be older and harder to reach, offering more security for birds avoiding disturbance caused by “catastrophic (lethal) herbivory” and in turn, increasing the probability of nest occurrence (Introduction, paragraph 2). The probability of nest occurrence may differ depending on the aggressive tendencies (or lack thereof) of the

ant species that has colonized the tree. More aggression to protect the tree may benefit birds' nests from herbivory disturbance, but it may also put the birds and their offspring at risk of attack by hostile ants (Introduction, paragraph 3).

d. Table of models

Model number	Tree height	Ant species	Model description
1	x	x	model with interaction term
2	x	x	model w/o interaction term

e. Run the models

```
# model 1: tree height and ant species w/ interaction
model1 <- glm(
  case_control ~ height_cm * ant_species,
  data = nests_clean,
  family = "binomial"
)

# model 2: tree height and ant species w/o interaction
model2 <- glm(
  case_control ~ height_cm + ant_species,
  data = nests_clean,
  family = "binomial"
)
```

f. Select the best model

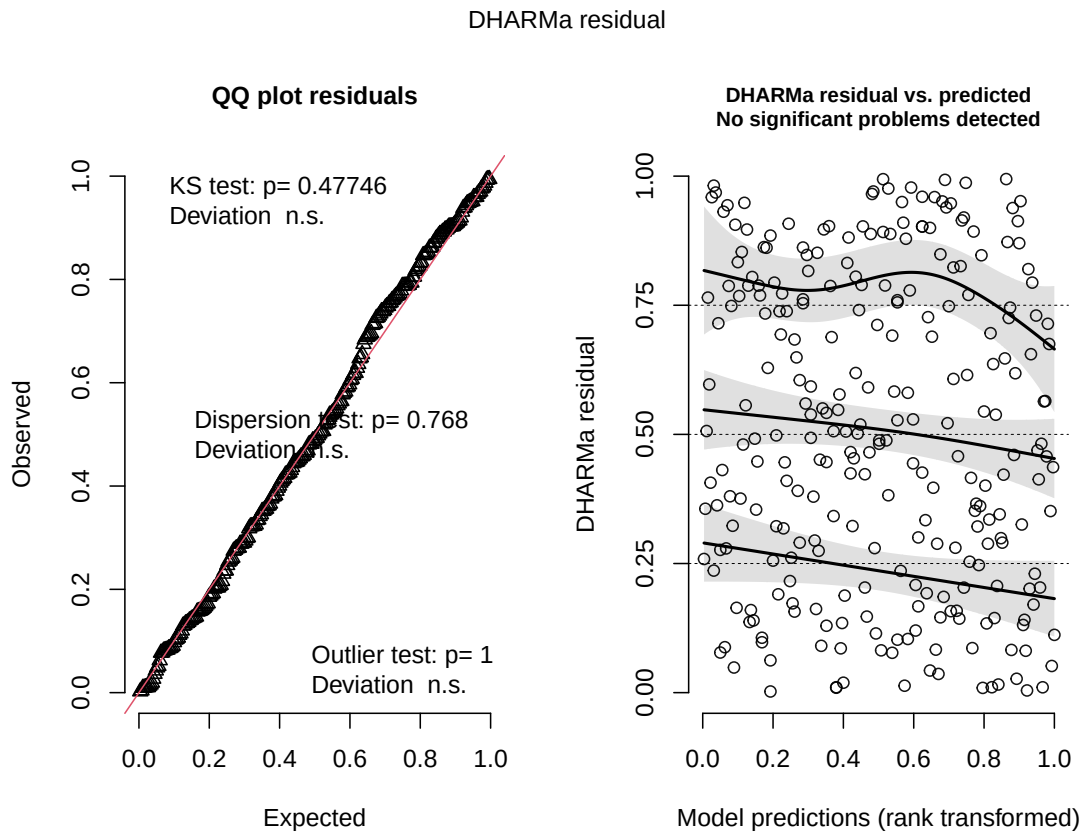
```
# used AIC to choose the best model
AICc(
  model1,
  model2
)
```

```
      df      AICc
model1  6 238.1236
model2  4 258.8940
```

The best model as determined by Akaike's Information Criterion (AIC) is the one that includes tree height and water treatment (with interaction terms) as predictors.

g. Check the diagnostics

```
# simulated residuals
plot(
  simulateResiduals(model1)
)
```



h. Visualize the model predictions

i. Write a caption for your figure

j. Calculate model predictions

k. Interpret your results

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

In the affective visualization from Homework 3, each observation is represented by an icon, resulting in an array of 30 icons. The exploratory visualizations I made in Homework 2 were plotted so that the response variable (mean reaction time in milliseconds) would be on the y-axes and the predictor variables (exercise duration in minutes for the scatter plot and exercise type for the strip plot) would be on the x-axes. Observations were denoted with points (in the strip plot, points were assigned unique colors by exercise type).

Between all my visualizations, a common feature they share is a round, circular representation for observations. In the exploratory visualizations, these are the plotted points, while the affective visualization is made up of round volleyball and bike tire icons. Other than visual elements, the plots share the same response variable (mean reaction time in milliseconds). The strip plot and affective visualization both present information framed by the categories of exercise type.

Starting with exploratory visualizations, the strip plot displays a wide range of response values for biking and a more narrow range for volleyball (due to the lower observation count). The data collection incomplete at the time, it was hard to tell if there was a difference in central tendency for mean reaction time between the groups. The scatter plot was similarly difficult to draw concrete statements from due to low observation counts. Most points were concentrated in shorter durations around 7 minutes because I'd collected more data focused on biking (and biking took less time than volleyball sessions). Taking this into account, the negative trend shown in the plot can be understood as a very vague pattern. And in the affective visualization, volleyball and biking appeared to be the most frequent activities based on the number of icons, while volleyball activities involved the most social interaction. Patterns are different between visualizations because I had attempted to highlight a different finding through each figure.

The feedback I received during week 9 in workshop were mainly focused on making an artistic (affective) visualization because at the time I did not have one. I was also encouraged to record more data. My group's comments motivated me to create a visualization that was easy to understand and that would clearly communicate my findings. With their suggestions in

mind, I went with an array of icons that would signal different factors such as socialization and hydration with additional colors and smaller icons.

b. Designing an analysis

Based on my response variable (mean reaction time in milliseconds) and chosen predictor value (exercise type), a Mann-Whitney U test can be used to analyze my data in addition to answering my question: Does my mean reaction time differ across different kinds of exercise? This analysis will compare the central tendencies of a continuous variable (mean reaction time) between two categorical variables (exercise type).

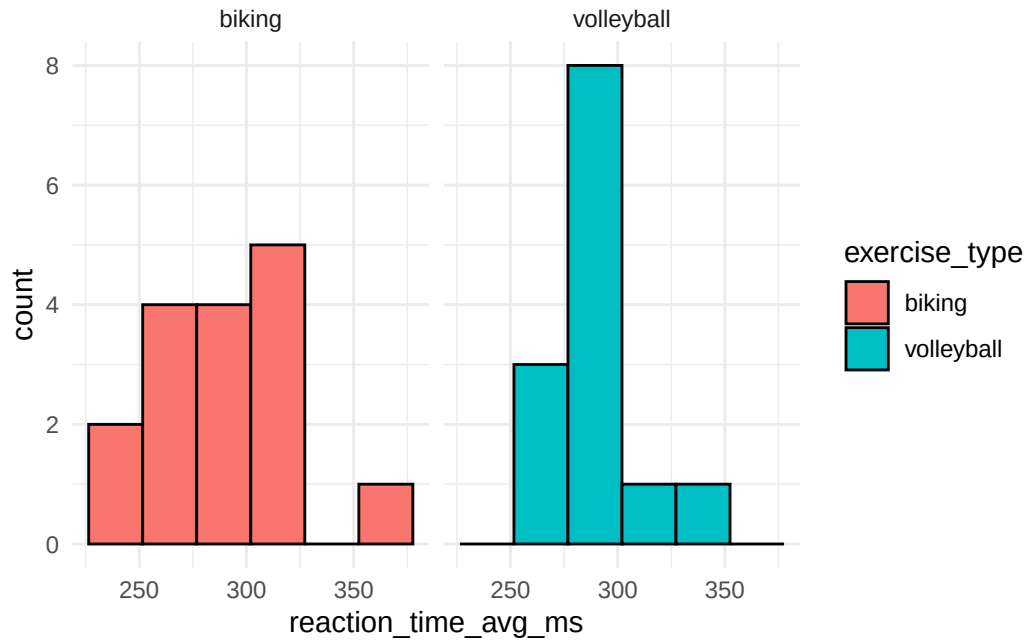
c. Check any assumptions and run your analysis

Cleaning data

```
# created new object from rxntime data frame
rxntime_clean <- rxntime |>
  # cleaned names
  clean_names() |>
  # filtered exercise types to exclude walking (insufficient data)
  filter(exercise_type != "walking")
```

Checking assumptions - Assessing normality with histograms

```
ggplot(data = rxntime_clean,
       mapping = aes(x = reaction_time_avg_ms,
                     fill = exercise_type)) +
  geom_histogram(bins = 6,
                color = "black") +
  facet_wrap(~ exercise_type) +
  theme_minimal()
```



Distributions are similarly right-skewed and data were randomly sampled, so a Mann-Whitney U test is appropriate for this data.

Running the analysis

```
# Mann-Whitney U test using clean data frame
wilcox.test(
  reaction_time_avg_ms ~ exercise_type,
  data = rxntime_clean
)
```

Wilcoxon rank sum test with continuity correction

```
data: reaction_time_avg_ms by exercise_type
W = 109.5, p-value = 0.8264
alternative hypothesis: true location shift is not equal to 0
```

Calculating effect size

```
# calculated effect size using clean data frame
cliff.delta(
  reaction_time_avg_ms ~ exercise_type,
  data = rxntime_clean
)
```

Cliff's Delta

```
delta estimate: 0.05288462 (negligible)
95 percent confidence interval:
      lower      upper
-0.3824587  0.4690131
```

d. Create a visualization

Cleaning data

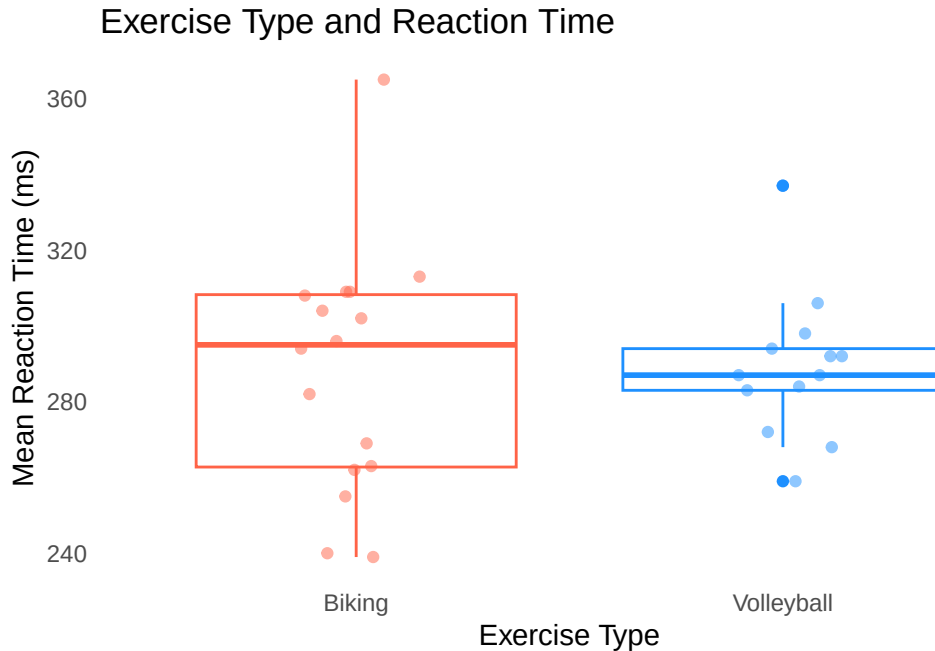
```
# created a new object from rxntime
rxntime_box <- rxntime |>
# cleaned column names
clean_names() |>
filter(exercise_type != "walking") |>
# selected columns of interest
select(exercise_type, reaction_time_avg_ms) |>
# capitalized existing abbreviations
mutate(exercise_type = case_when(
  exercise_type == "volleyball" ~ "Volleyball",
  exercise_type == "biking"     ~ "Biking")) |>
# removed missing values
drop_na()
```

Visualization

```

# base layer: ggplot
ggplot(data = rxn_time_box,
      # x-axis: exercise type
      # y-axis: mean reaction time
      # color by exercise type
      mapping = aes(x = exercise_type,
                    y = reaction_time_avg_ms,
                    color = exercise_type)) +
# first layer: boxplot
geom_boxplot() +
# second layer: data points and jitter
geom_jitter(width = 0.15, # range of horizontal jitter
            height = 0,    # no vertical jitter
            alpha = 0.5) + # lowered opacity of raw data
# relabeled x and y axes, added a title
labs(title = "Exercise Type and Reaction Time",
      x = "Exercise Type",
      y = "Mean Reaction Time (ms)") +
# custom theme
theme_minimal() +
# removed legend
theme(legend.position = "none") +
# custom colors
scale_color_manual(values = c("Biking" = "tomato",
                              "Volleyball" = "dodgerblue")) +
# removed grid and axis lines
theme(panel.grid = element_blank(),
      axis.line = element_blank())

```



e. Write a caption

Figure 1. Mean reaction times across three exercise types. Box/ jitter plots of mean reaction times (ms) for three exercise types (red: biking, blue: volleyball). Interquartile ranges fall around 260-310 milliseconds and median values are 295 and 287 milliseconds for biking and volleyball, respectively. Points represent individual observations. Data are from my personal observations throughout the spring quarter.

f. Write about your results

We found no statistically significant difference in median reaction times between volleyball (median = 287 ms, $n = 13$) and biking (median = 295 ms, $n = 16$) exercises (Wilcoxon rank-sum test, $W = 109.5$, $p = 0.83$, $\alpha = 0.05$). There was a small (Cliff's $\delta = 0.05$) difference in median reaction times between exercise types.

g. Sharing your affective visualization

I was there that day Cool visualizations Everywhere, they were